Notes from Ben, not part of the talk

- Side chain: now explained on slide 2 in one clause (Ben, 14 October). The whole deck was revised on 14 October along the four facts of slide 6, see `WORK-SPLIT.md`.

Highly accurate protein structure prediction with AlphaFold

Paper by Jumper et al., *Nature* 596 (2021)

Evita Bosch Pokunevich Ben Bullinger

Machine Learning for Genomics (263-5351-00L), ETH Zürich, Fall 2026

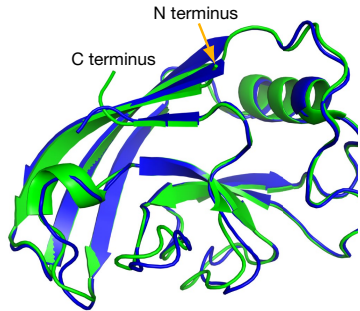
Outline

1. The problem
2. The benchmark
3. The network
   - The data
   - The network, in detail
4. Training
5. What makes it work
6. Conclusion
7. Questions

Sequence in, structure out

SIFSYITESTGTPS…

sequence of amino acids. Each has the same backbone and its own side chain



3D coordinates of all heavy atoms
Prediction in blue, experiment in green

An experiment takes months to years

Billions of sequences are known

The problem is more than 50 years old

Sequence and structure are the CASP14 target T1049, PDB 6Y4F. The structure is Fig. 1b.

The Benchmark: CASP14

CASP14: Critical Assessment of protein Structure Prediction

May-July 2020

Blind test - experimental structures are not revealed until the end of experiment.

Å (angstrom): a unit of length equal to 10^{-10} meters.

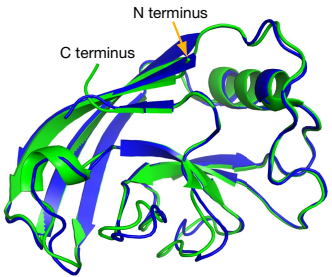
Smaller RMSD means a closer structural match.

Show the average prediction hist from the paper

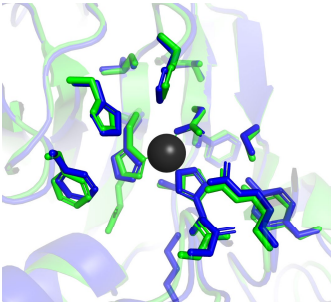
The median backbone error is below one atom width

Redrawn from Fig. 1a. The error is the Cα r.m.s.d.$_{95}$ on 87 domains, with 95% confidence intervals.

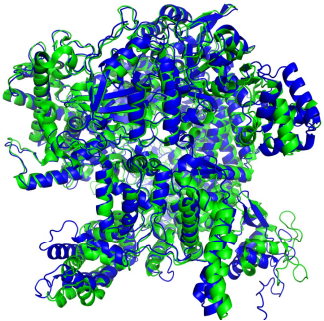
Predictions on top of experiment – delete?



Backbone



Side chains

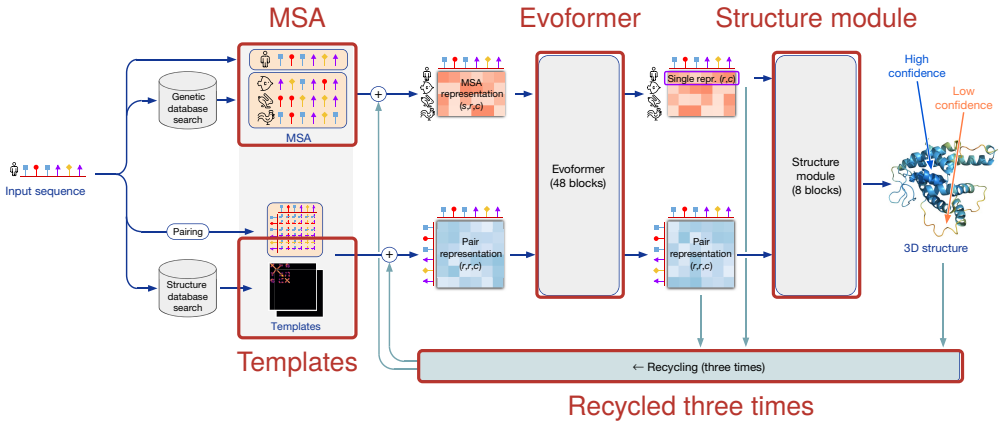


A 2,180-residue chain

AlphaFold experiment

Prediction and experiment overlap, also in the side chains

One pipeline, end to end



Coordinates come out of the network itself

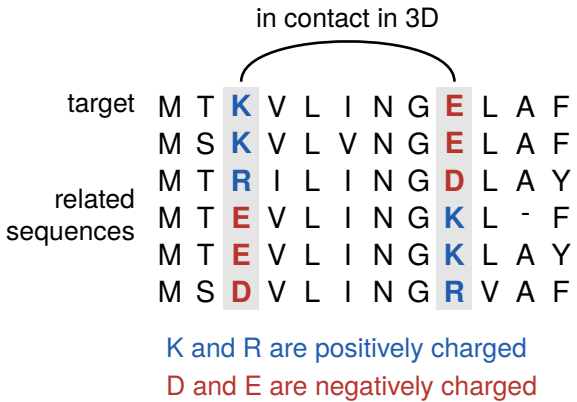
Four facts about proteins, four design choices

		Fact	Design choice in AlphaFold 2
1		An MSA is a set of sequences. A protein is a chain of residues.	Attention over the sequences themselves, along the rows and the columns.
2		Distances in 3D must close triangles.	The pairs are the edges of a residue graph, updated through triangles.
3		A structure is the same after any rotation or translation.	One frame per residue. Attention uses only distances in the frames.
4		A protein and its mirror image are different molecules.	The loss compares atoms in local frames. A mirror image is penalised.

Removing any one of the four costs accuracy. The ablations measure how much.

The input

The input: MSA representation



A row is a related protein

A column is an aligned position

Contacts may change together

The network reads the sequences themselves

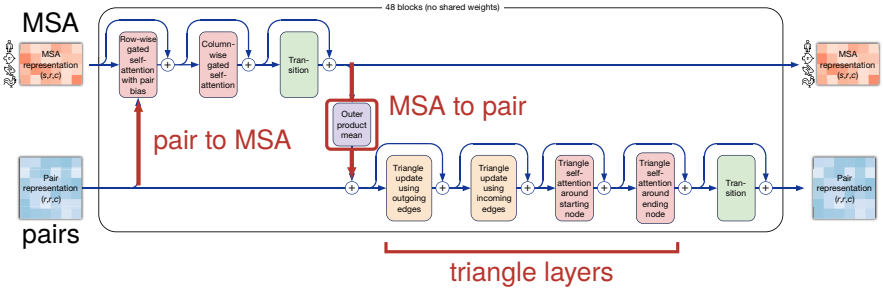
D. S. Marks et al., *PLoS ONE* 6, 2011. The alignment is an illustration, not data.

The input: Pair representation

Evoformer block



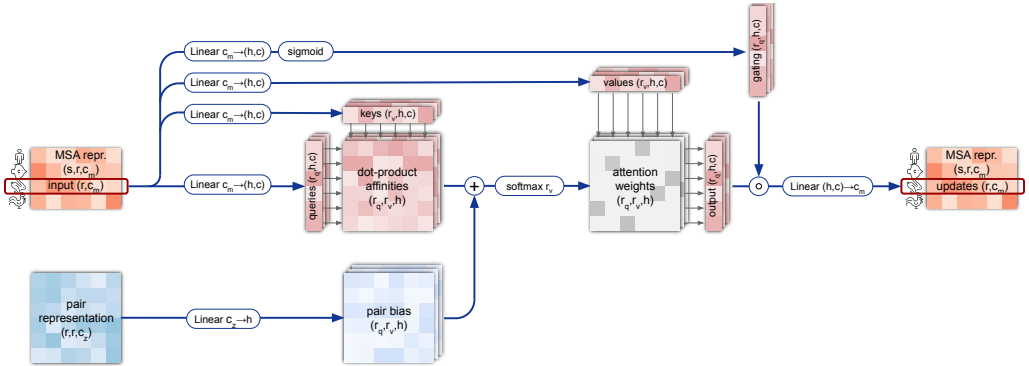
1. The sequences in an MSA are a set. The residues are a chain.



Attention along the rows and the columns. MSA and pair representation exchange information

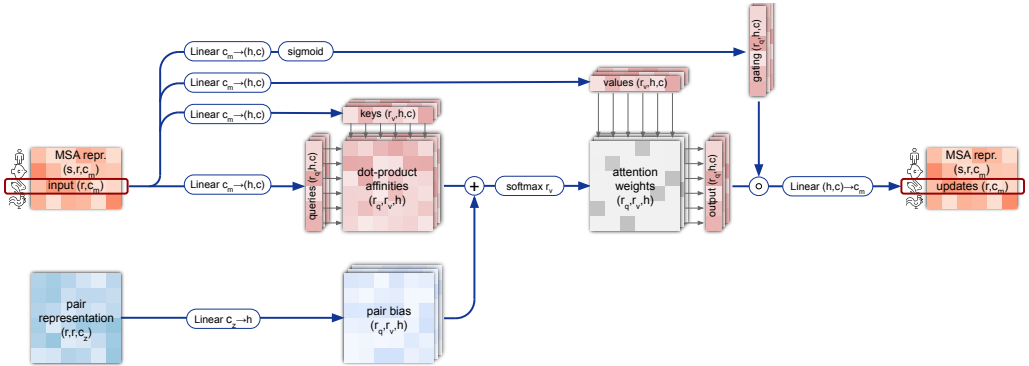
The paper calls this graph inference. The edges are residues in proximity.

Evoformer: MSA representation, row col



Coordinates come out of the network itself

Evoformer: MSA representation, trans out

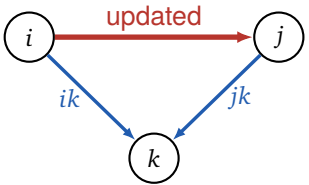


Coordinates come out of the network itself

Evoformer: Pair representation

2. Distances in 3D must close triangles.

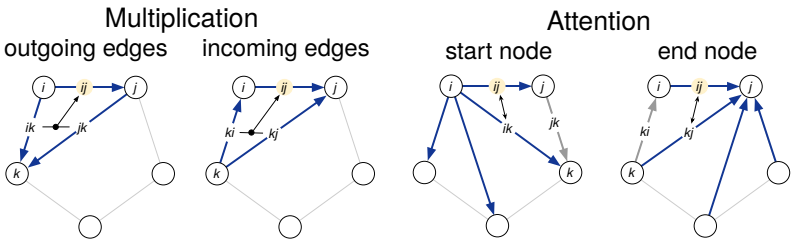




The pair representation is the edge set of the graph (Fig. 3b)

Edge ij is updated from the edges ik and jk

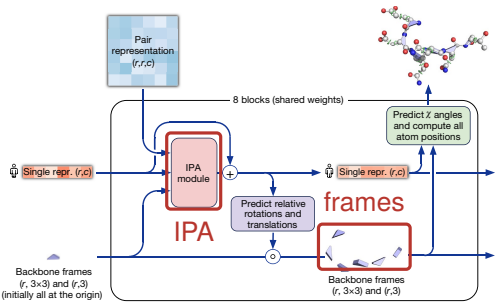
The triangle inequality acts as an inductive bias

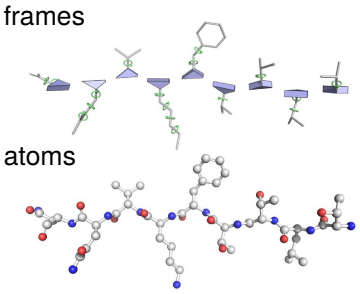


Structure module: residues as rigid bodies



3. A structure is the same after any rotation or translation.





A frame is one rotation and one translation per residue

All frames are updated at once

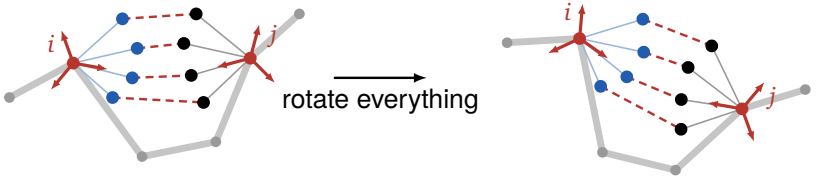
Nothing forces neighbouring residues to stay connected

Invariant point attention



3. A structure is the same after any rotation or translation.

$$a_{ij} = \mathrm{softmax}_j\Big(\underbrace{\boldsymbol{q}_i^\top \boldsymbol{k}_j}_{\text{usual attention}} + \underbrace{b_{ij}}_{\text{pair bias}} - \underbrace{\textstyle\sum_p \left\| T_i \circ \vec{\boldsymbol{q}}_{ip} - T_j \circ \vec{\boldsymbol{k}}_{jp} \right\|^2}_{\text{distances between 3D points}} \Big)$$



Points are placed in the frames of i and j

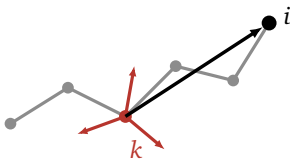
The distances stay the same

The weights are invariant to global rotation and translation

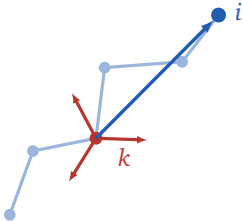
The formula is Algorithm 22 of the supplement for one head, without scaling constants.

FAPE: a loss over local frames

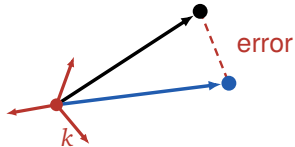
4. A protein and its mirror image are different molecules.



experiment



prediction



both, seen from frame k

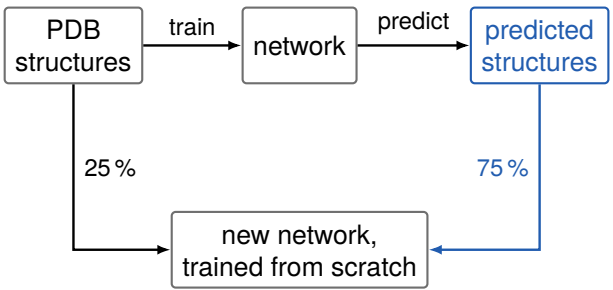
$$\text{FAPE} = \frac{1}{Z} \underset{k,i}{\text{mean}} \min\left(d_{\text{clamp}}, \left\|(\boldsymbol{R}_k, \boldsymbol{t}_k)^{-1}\boldsymbol{x}_i - (\boldsymbol{R}_k^{\text{true}}, \boldsymbol{t}_k^{\text{true}})^{-1}\boldsymbol{x}_i^{\text{true}}\right\|\right)$$

A rotation changes nothing. A mirror image is penalised.

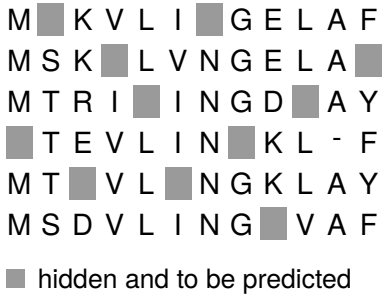
Supplementary Information, Algorithm 28, with $Z = d_{\text{clamp}} = 10\,\text{Å}$. The drawing is an illustration.

Trained beyond the PDB

Self-distillation adds 350,000 structures



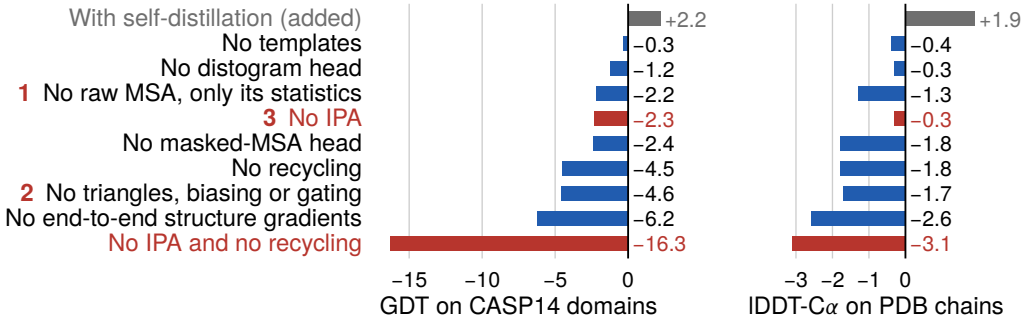
A masked-MSA loss is added



Training uses sequences without structures

Self-distillation follows Q. Xie et al., 2020. Masking follows BERT, J. Devlin et al., 2019.

Which pieces earn their accuracy

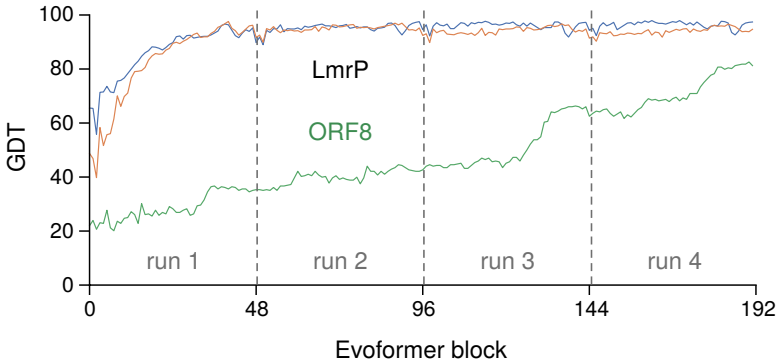


4 With a mirror-blind loss instead of FAPE, the model often builds the mirror image

Each of the four choices costs accuracy on CASP14 when it is removed

Redrawn from Fig. 4a, the mean change against the baseline over the training runs of a row. Line 4: Suppl. Fig. 9.

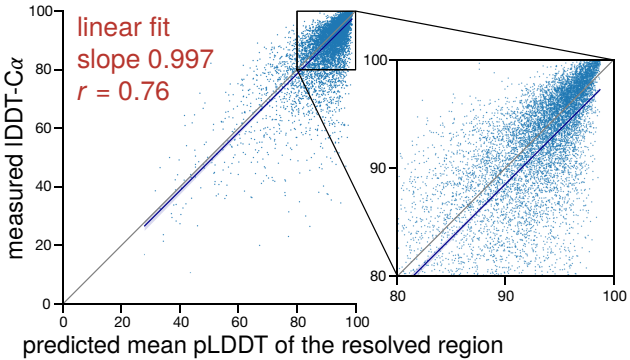
The structure forms inside the trunk



The trunk is frozen and every block gets a structure module

Some proteins are solved early and others late

The model knows when it is right

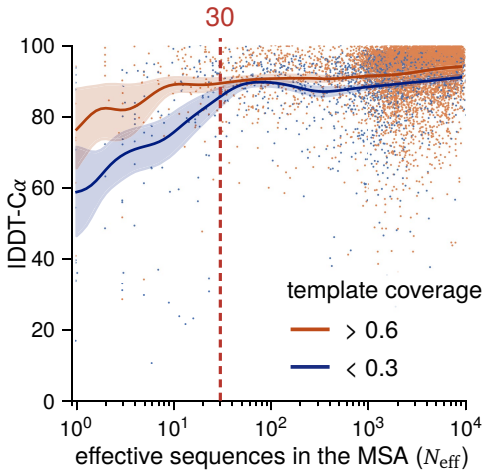


pLDDT is the predicted lDDT-Cα per residue

Predictions need a well-calibrated confidence

K. Tunyasuvunakool et al., *Nature* 596, 2021.

Where it still fails



Accuracy falls below about 30 effective sequences

AlphaFold is weaker for chains shaped by other chains

AlphaFold needs enough related sequences

What followed

2021 — RoseTTAFold appears online the same day

98.5% of human proteins are predicted

AlphaFold-Multimer predicts complexes

2022 — ESMFold uses a language model and no MSA

2024 — AlphaFold 3 is one model for proteins, DNA, RNA and ligands

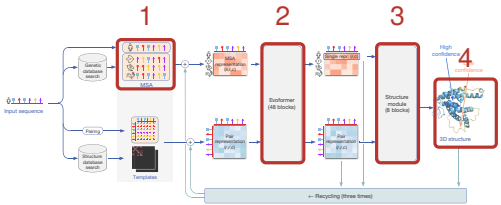
address the limits

Baker, Hassabis and Jumper receive the Nobel Prize in Chemistry

AlphaFold 3 is more accurate than specialised tools, with one exception

The first authors are Baek 2021, Tunyasuvunakool 2021, Evans 2021, Lin 2022 and Abramson 2024.

Summary



		GDT lost
1	attention over sequences	−2.2
2	triangles on the pairs	−4.6
3	frames and IPA	−2.3
4	FAPE in local frames	mirrored

Evolution and geometry are trained end to end, in about one GPU minute per model

The median backbone error is below one atom width

J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” *Nature* 596, 583–589 (2021). GDT lost on CASP14 when removed, Fig. 4a and Suppl. Fig. 9.

Exam question

Placeholder. The question from the course team arrives about seven days before the talk.

Our answer.

Questions for the audience

Placeholder. At least two questions, then a moderated discussion.

1. First question.
2. Second question.

Why is the attention invariant to a rigid motion?

SI 1.8.2, Algorithm 22, equations 3 to 6



The only geometry in the weights is the distance between points that i and j place in their own frames

$$\left\|(T_{\text{global}} \circ T_i) \circ \vec{\boldsymbol{q}}_i - (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{k}}_j\right\|^2$$
$$= \left\|T_{\text{global}} \circ (T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j)\right\|^2 = \left\|T_i \circ \vec{\boldsymbol{q}}_i - T_j \circ \vec{\boldsymbol{k}}_j\right\|^2$$

moving both points together leaves their distance unchanged

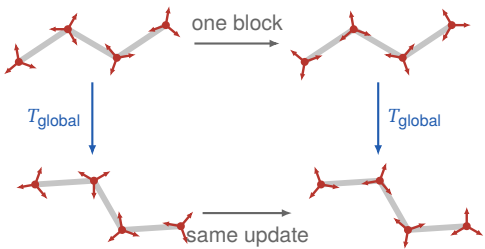
$$(T_{\text{global}} \circ T_i)^{-1} \circ \textstyle\sum_j a_{ij}\, (T_{\text{global}} \circ T_j) \circ \vec{\boldsymbol{v}}_j$$
$$= T_i^{-1} \circ T_{\text{global}}^{-1} \circ T_{\text{global}} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j = T_i^{-1} \circ \textstyle\sum_j a_{ij}\, T_j \circ \vec{\boldsymbol{v}}_j$$

the output points are read back in frame i, and T_{global} meets its inverse. Exact, not learned.

Weights and outputs do not depend on where the structure sits

Why is the structure module equivariant?

SI 1.8.2 and 1.8.3, Algorithms 20 and 23



The update of a frame is predicted from $\boldsymbol{s}_i$, which is invariant, and composed in the frame's own coordinates

$(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$

moving the input first or the output last gives the same frames

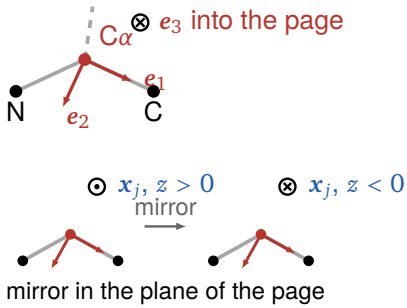
"Rigid motion" in the supplement means proper rotation and translation, not reflection. In group language: SE(3).

All frames start as the identity, so the global frame of the output is arbitrary, and the loss has to be invariant to it

Exact SE(3) equivariance. Reflections are excluded on purpose.

Where does the handedness come from?

SI 1.8.1, 1.9.3 and 1.8.4, Algorithm 21, equations 11 to 17



A frame is built from N, Cα and C by Gram–Schmidt, with $\boldsymbol{e}_3 = \boldsymbol{e}_1 \times \boldsymbol{e}_2$. It is always right-handed.

Reflect all atoms. The frames do not reflect with them. In local coordinates exactly z flips:

$$T_i^{-1} \circ \boldsymbol{x}_j \;\mapsto\; \mathrm{diag}(1, 1, -1)\,(T_i^{-1} \circ \boldsymbol{x}_j)$$

$$\mathrm{FAPE}(X, X_{\mathrm{mirror}}) = 2 \sum_{ij} \left|(T_i^{-1} \circ \boldsymbol{x}_j)_z\right|$$

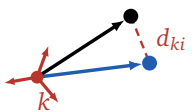
large unless the structure is flat

Trained with dRMSD, a loss blind to reflection, the model builds the mirror image in a large share of cases (Suppl. Fig. 9)

The loss sees the hand, the data teach it. Side chains are always built L.

What does FAPE measure, and when is it zero?

SI 1.9.2, 1.9.4 and 1.9.5, Algorithm 28, equations 18 to 30



seen from frame k

Every atom i in every frame k, in local coordinates: $N_{\text{frames}} \times N_{\text{atoms}}$ terms, each clamped at 10 Å and divided by 10 Å

Invariant to a rigid motion of the prediction, and of the truth, by the same cancellation as in IPA

Lower bound: FAPE is at least the RMSD after the best rigid alignment, whatever the frames (eqs. 18 to 22). So it is zero only if the two point sets differ by a proper rigid motion.

Non-negative, symmetric, triangle inequality: a pseudometric on structures (eqs. 23 to 30)

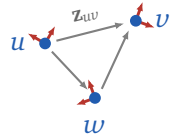
Local geometry in every frame: blind to placement, sensitive to the hand

Is AlphaFold 2 a graph neural network?

Formalism of Bronstein, Bruna, Cohen and Veličković, 2021

$$\mathbf{h}_u = \phi\Big(\mathbf{x}_u, \bigoplus_{v \in \mathcal{N}_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv})\Big)$$

message ψ, aggregation $\bigoplus$, update ϕ. Nodes: residues, $\mathbf{s}_u$ and frame T_u. Edges: $\mathbf{z}_{uv}$.



Block	message ψ	into
MSA row attention	$\psi(\mathbf{m}_{su}, \mathbf{m}_{sv}, \mathbf{z}_{uv})$, the edge biases the weight	node u
outer product mean	$\psi(\mathbf{m}_{\cdot u}, \mathbf{m}_{\cdot v})$, averaged over the sequences	edge uv
triangle updates	$\psi(\mathbf{z}_{uw}, \mathbf{z}_{wv})$ over every third node w	edge uv
IPA	$\psi(\mathbf{s}_u, \mathbf{s}_v, \mathbf{z}_{uv}, \lVert T_u \circ \vec{\mathbf{q}}_u - T_v \circ \vec{\mathbf{k}}_v \rVert)$, values moved by $T_u^{-1} \circ T_v$	node u
backbone update	none, ϕ alone	frame T_u

Triangles are two-hop messages. They make the edges consistent with one structure.

Geometry enters only through distances. A frame is a gauge fixed by its atoms.

What does each block compute?

Supplement, Algorithms 6 to 32

	Update	Message	SI
Evoformer, 48 blocks			
1	$\mathbf{m} \leftarrow \mathbf{m} + \text{RowAttention}(\mathbf{m} \mid \mathbf{z})$	node → node, edge biases the weight	Alg. 7
2	$\mathbf{m} \leftarrow \mathbf{m} + \text{ColumnAttention}(\mathbf{m})$	across the sequences at one node	Alg. 8
3	$\mathbf{m} \leftarrow \mathbf{m} + \text{MLP}(\mathbf{m})$	per node	Alg. 9
4	$\mathbf{z} \leftarrow \mathbf{z} + \text{OuterProductMean}(\mathbf{m})$	node pair → edge	Alg. 10
5	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleMultiplication}(\mathbf{z})$, out and in	edge pair → edge	Alg. 11, 12
6	$\mathbf{z} \leftarrow \mathbf{z} + \text{TriangleAttention}(\mathbf{z})$, start and end node	edge pair → edge	Alg. 13, 14
7	$\mathbf{z} \leftarrow \mathbf{z} + \text{MLP}(\mathbf{z})$	per edge	Alg. 15
Structure module, 8 blocks with shared weights. At the start $\mathbf{s}_u = \mathbf{m}_{1u}$ and T_u is the identity.			
8	$\mathbf{s} \leftarrow \mathbf{s} + \text{IPA}(\mathbf{s}, \mathbf{z}, T)$	node → node, values moved by $T_u^{-1} \circ T_v$	Alg. 22
9	$\mathbf{s} \leftarrow \mathbf{s} + \text{MLP}(\mathbf{s})$	per node	Alg. 20
10	$T_u \leftarrow T_u \circ \text{BackboneUpdate}(\mathbf{s}_u)$	per node, in its own frame	Alg. 23
Recycling, 4 cycles			
11	$\mathbf{z} \leftarrow \mathbf{z} + \text{Embed}(d_{uv})$, C$\beta$ distances from the frames	frames → edges	Alg. 32

Every line is $\mathbf{h}_u = \phi(\mathbf{x}_u, \bigoplus_v \psi)$ on nodes or on edges. Information moves in lines 1, 4, 5, 6, 8 and 11.

What exactly did each ablation remove?

SI 1.13.1 to 1.13.3

Row of Fig. 4a	What the model has instead
No raw MSA	the profile and the pairwise frequencies of the MSA, in one row, no column attention
No triangles, biasing or gating	axial attention on the pairs with a relative position bias, no gating
No IPA	frames from one linear projection of the single representation, no iteration
No recycling	a single pass through the network
No end-to-end structure gradients	the structure losses stop before the Evoformer, which learns from the auxiliary losses only
No templates	no template stack, four more MSA clusters instead

Baseline: the full model without self-distillation. Three or four seeds, not re-tuned

Not additive: no IPA costs 2.3, no recycling 4.5, both together 16.3 GDT

How does the network predict its own accuracy?

SI 1.9.6 and 1.9.7, Algorithm 29

pLDDT: a small network on the final single representation predicts, per residue, a distribution over 50 bins of lDDT-Cα. It is trained with cross-entropy against the lDDT-Cα of the prediction itself, on structures resolved between 0.1 and 3 Å

The reported pLDDT is the expected value of that distribution. The pLDDT of a chain is the mean over its residues

pTM: a pairwise head predicts the distribution of the aligned error of every residue pair. The predicted TM-score follows from these distributions

Confidence is a prediction, trained on the model's own errors

How was it trained?

Methods, “Training regimen”, SI 1.3 and 1.11.1, Table 4

Data: PDB structures released before 30 April 2018, and about 350,000 predicted structures of Uniclust30 sequences without a structure, filtered by confidence, 75 % of the samples

Stage 1: crops of 256 residues, about 10 million samples, about seven days on 128 TPU v3 cores

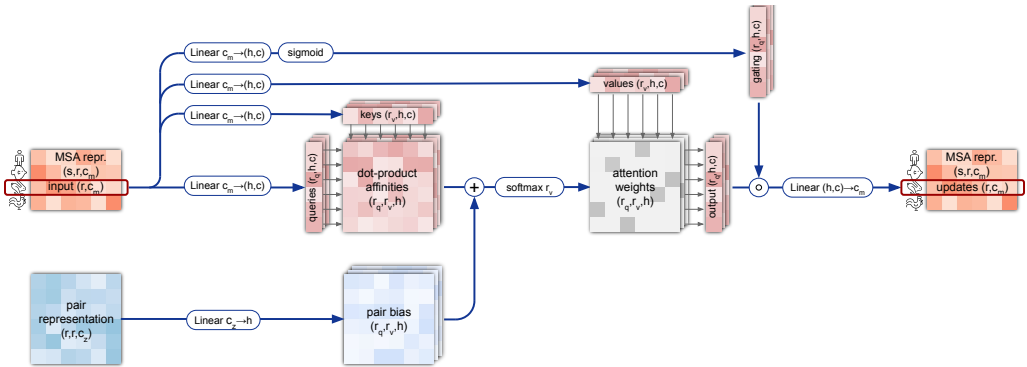
Stage 2: crops of 384 residues, the structural violation loss switched on, about 1.5 million samples, about four days

Losses: FAPE, torsion angles, distogram, masked MSA, pLDDT, experimentally resolved, violations

Eleven days on 128 TPU cores per model. Inference takes GPU minutes to hours.

MSA row attention with pair bias

Supplementary Fig. 2, Algorithm 7



Supplementary Fig. 2. Every sequence of the MSA attends along itself, the pair representation biases the weights